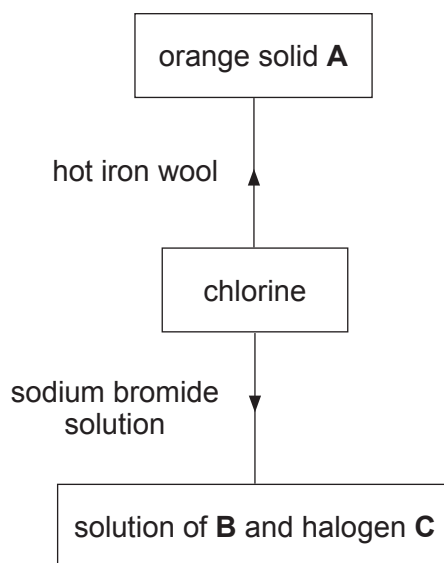


5. (a) The diagram shows some reactions of chlorine.



- (i) Give the chemical **formulae** for substances **A**, **B** and **C**. [3]

A

B

C

- (ii) State, giving a reason, the **main** safety precaution that must be taken when carrying out these reactions. [1]

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- (iii) State and explain how the observations for the reaction with sodium bromide would be different if the chlorine were replaced with iodine. [3]

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- (b) Chlorine can form a number of compounds by reacting with oxygen gas. One of these is used in water treatment.

A sample of this compound contains 0.71 g of chlorine and 0.64 g of oxygen. Calculate its simplest formula. [3]

Simplest formula

- (c) Silver nitrate solution is used to detect the presence of halide ions in solution.

- (i) The equation below represents the reaction between silver nitrate solution and potassium bromide solution.



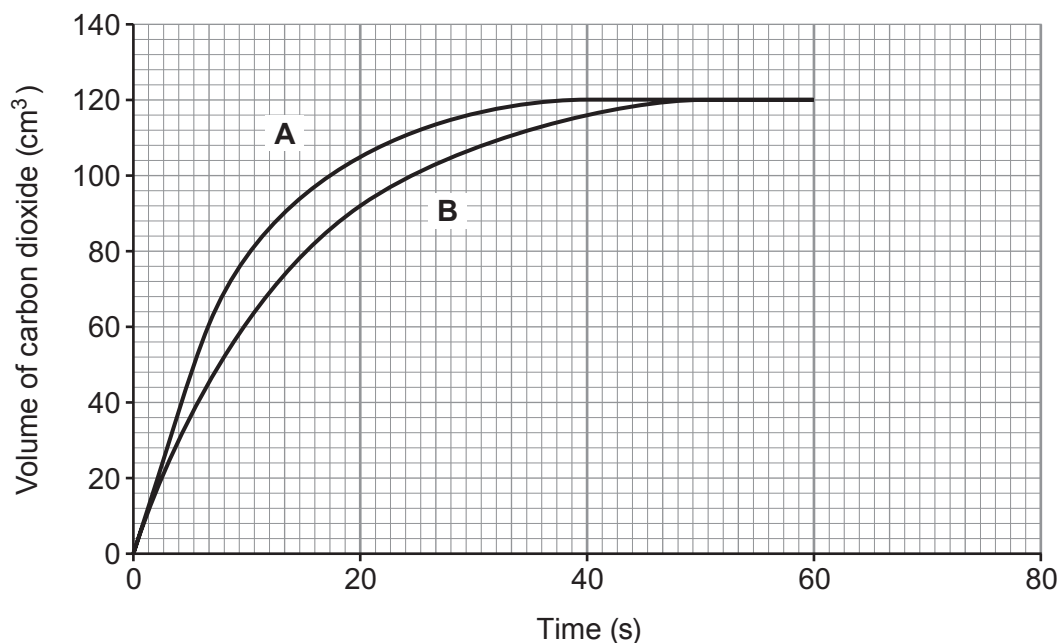
Write the **ionic** equation for the reaction. **Include state symbols.** [2]

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- (ii) Write the balanced **symbol** equation for the reaction between calcium iodide solution and silver nitrate solution. [3]

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5. The rate of reaction between hydrochloric acid and calcium carbonate was studied. Two experiments, **A** and **B**, were carried out. The same concentration of acid and the same mass of calcium carbonate were used, with the acid in excess each time. The volume of gas produced was measured for 60 seconds. The results of these experiments are shown in the graphs.



- (a) Using the particle theory, state and explain **two** factors that may be responsible for the higher rate seen in experiment **A**. [4]

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- (b) State how the graphs show that the same mass of calcium carbonate was used in both experiments. [1]

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Examiner
only

5. (a) A bottle contains a mixture of liquids **E** and **F**. Liquid **E** has a boiling point of 57 °C and liquid **F** has a boiling point of 95 °C.

Describe the process of distillation and explain why it can be used to separate these liquids. [3]

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- (b) One molecule of liquid **E** contains **two** oxygen atoms. The percentage by mass of oxygen in liquid **E** is 43.2 %.

Use the following equation to calculate the relative molecular mass (M_r) of liquid **E**. [2]

$$\frac{\text{mass of oxygen}}{M_r} \times 100 = 43.2$$

$$A_r(\text{O}) = 16$$

$M_r =$

5



- State the difference between the composition of temporary hard water and permanent hard water. Describe a method to distinguish between them in the laboratory. Explain how this method works.

[6 QER]

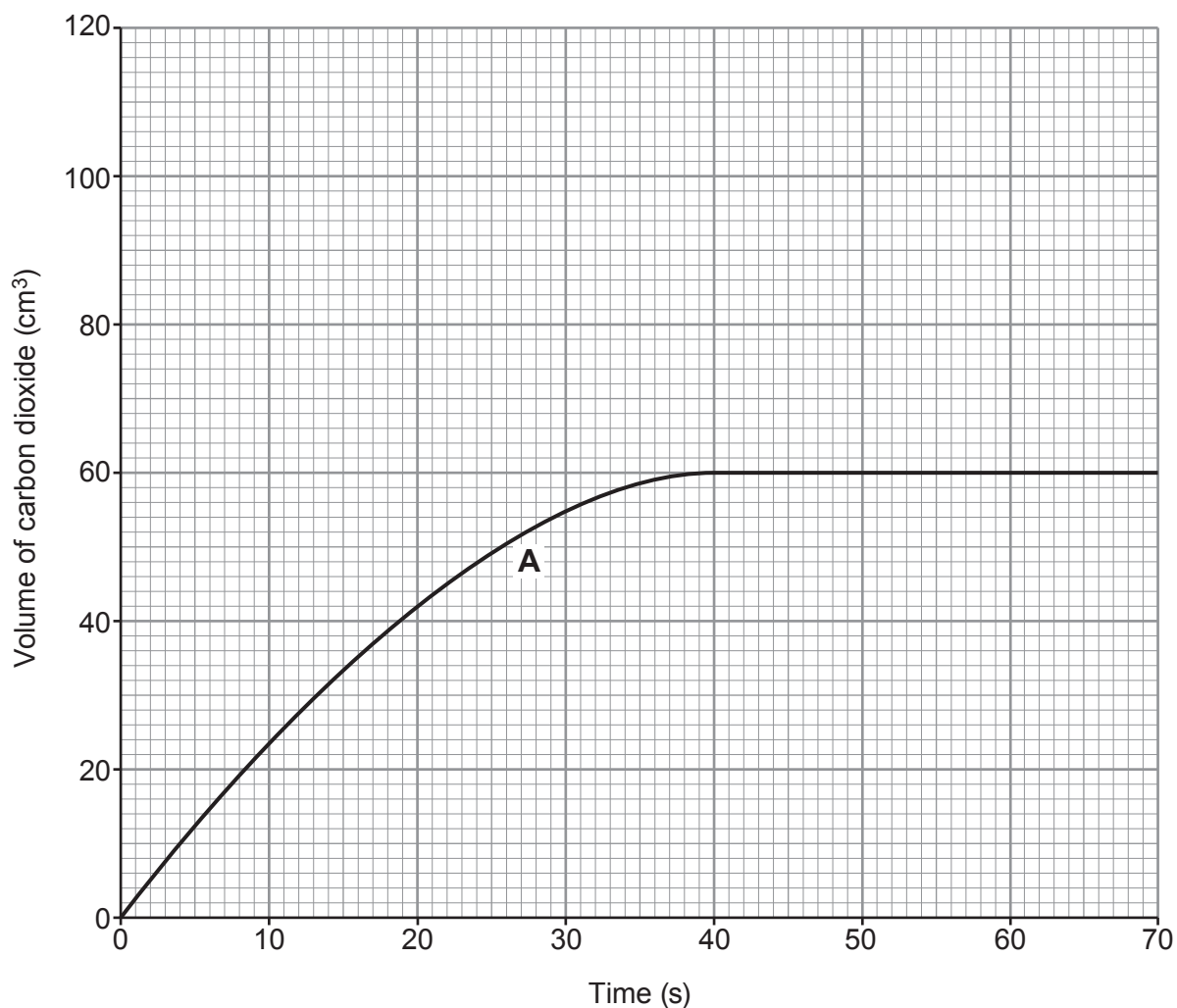
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5. Marble chips (calcium carbonate) react with dilute hydrochloric acid to produce carbon dioxide gas.

Graph **A** shows the volume of carbon dioxide produced during the reaction between 0.25 g of marble chips and **excess** dilute hydrochloric acid at 30 °C.



- (a) State why no more marble chips remain when the reaction stops. [1]

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- (b) The original experiment was repeated at 60 °C.

On the grid draw the curve you would expect to see for the production of carbon dioxide at this temperature. [1]



- (c) The equation for the reaction between calcium carbonate and dilute hydrochloric acid is as follows.



Calculate the mass of calcium chloride produced when 7.8 g of calcium carbonate is reacted with excess acid. [2]

$$M_r(\text{CaCO}_3) = 100$$

$$M_r(\text{CaCl}_2) = 111$$

Mass = g

- (d) When the reaction was repeated with a different mass of calcium carbonate, it was found that 14.3 g of calcium chloride formed. This represents a yield of 53.7%.

Calculate the maximum mass of calcium chloride that could have been formed. [2]

Maximum mass = g

- (e) Catalysts can be added to some chemical reactions in order to increase their rate.

Explain how a catalyst increases the rate of a chemical reaction. [2]

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